

SOUTHWEST WYOMING, WY, USA

WMO: 725744

Lat: 41.595N

Lon: 109.053W

Elev: 2055

StdP: 78.95

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24027

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.3	-17.2	-23.5	0.6	-19.3	-20.7	0.8	-14.8	16.5	-2.8	15.4	-3.4	3.7	60	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	31.3	12.9	29.9	12.4	28.5	12.1	14.8	25.6	14.0	24.9	13.3	24.5	5.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.5	10.9	16.7	10.2	10.0	16.5	8.9	9.1	16.4	48.0	25.3	45.5	24.6	43.4	24.4	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.8	11.6	DB	-24.9	33.6	3.0	1.1	-27.1	34.4	-28.8	35.0	-30.5	35.7	-32.7	36.4	(4)
(5)			WB	-25.2	16.1	2.9	1.1	-27.3	16.9	-29.1	17.5	-30.7	18.2	-32.9	19.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-5.4	-4.1	1.1	5.1	10.3	16.3	21.1	19.7	14.2	6.9	-0.1	-5.6	(6)	
	DBStd	10.40	5.46	5.37	5.03	4.70	4.72	4.28	2.55	2.65	4.49	5.16	5.47	5.56	(7)	
	HDD10.0	2288	477	394	277	157	56	6	0	0	17	119	303	482	(8)	
	HDD18.3	4432	736	627	534	398	250	85	7	17	131	354	552	741	(9)	
	CDD10.0	1083	0	0	1	9	65	195	345	299	144	23	1	0	(10)	
	CDD18.3	186	0	0	0	0	1	24	94	59	8	0	0	0	(11)	
	CDH23.3	2527	0	0	0	1	37	413	1171	748	154	3	0	0	(12)	
	CDH26.7	768	0	0	0	0	4	111	431	205	18	0	0	0	(13)	
	Wind	WSAvg	5.0	5.6	5.6	5.5	5.7	5.1	4.8	4.1	4.0	4.3	5.0	5.1	5.3	(14)
(15) Precipitation	PrecAvg	208	11	11	15	20	29	25	18	17	20	21	11	11	(15)	
	PrecMax	453	35	39	48	70	116	87	83	64	81	60	41	39	(16)	
	PrecMin	86	0	0	0	1	1	0	0	1	0	0	0	0	(17)	
	PrecStd	68	9	9	12	16	26	20	18	14	17	16	10	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	10.0	17.5	21.4	26.7	31.9	33.5	31.9	28.7	23.2	15.7	9.0	(19)	
		MCWB	1.9	3.1	5.9	7.9	10.7	12.6	14.2	13.3	12.0	9.2	5.5	2.4	(20)	
	2%	DB	5.2	7.0	14.1	18.7	24.0	29.6	32.0	30.2	26.8	20.4	12.8	5.8	(21)	
		MCWB	0.4	1.7	4.6	6.5	9.6	11.8	13.3	12.4	11.0	8.2	4.3	1.0	(22)	
	5%	DB	3.2	4.8	11.6	16.5	22.0	27.7	30.7	28.9	25.2	17.8	10.3	3.6	(23)	
		MCWB	-0.4	0.5	3.6	5.5	8.9	11.1	12.9	12.1	10.1	7.0	3.4	-0.1	(24)	
	10%	DB	1.8	3.0	8.9	14.0	19.6	25.8	29.2	27.6	23.2	15.5	7.9	1.9	(25)	
		MCWB	-1.2	-0.6	2.4	4.6	7.9	10.6	12.5	11.9	9.8	6.2	2.1	-0.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	3.6	6.9	8.5	11.9	14.0	16.0	15.6	14.1	10.4	6.5	3.3	(27)	
		MCDB	6.5	8.6	15.8	20.1	23.6	27.5	28.0	25.3	24.1	20.0	13.5	8.0	(28)	
	2%	WB	1.1	2.1	5.2	7.1	10.5	12.9	15.1	14.6	12.6	9.1	5.0	1.5	(29)	
		MCDB	4.1	6.2	12.8	16.8	21.2	25.5	26.2	24.8	21.8	17.7	11.5	4.8	(30)	
	5%	WB	0.0	0.9	3.9	6.0	9.5	12.0	14.3	13.8	11.7	8.0	3.8	0.3	(31)	
		MCDB	2.8	4.3	10.8	15.0	19.7	24.3	25.3	24.1	21.0	16.3	9.7	3.2	(32)	
	10%	WB	-1.0	-0.2	2.7	5.1	8.6	11.3	13.7	13.0	10.8	6.8	2.5	-0.8	(33)	
		MCDB	1.6	2.8	8.5	13.1	18.3	23.5	25.0	23.5	20.0	14.5	7.5	1.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.1	10.7	12.3	13.3	15.0	15.5	15.0	14.2	12.2	10.2	8.5	(35)	
		MCDBR	9.4	10.3	14.0	15.8	16.2	17.2	16.9	16.7	16.4	15.5	13.1	9.8	(36)	
	5% WB	MCWBR	6.3	6.7	7.1	7.3	6.6	6.7	6.1	6.2	6.8	7.3	7.2	6.5	(37)	
		MCWBR	8.7	9.5	13.2	14.5	14.8	15.7	14.6	14.1	13.7	14.0	12.1	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.228	0.235	0.250	0.279	0.297	0.298	0.327	0.321	0.286	0.254	0.234	0.228	(40)	
		taud	2.584	2.595	2.576	2.485	2.480	2.495	2.433	2.447	2.543	2.629	2.613	2.587	(41)	
	Ebn at Noon		966	1004	1015	995	977	972	939	939	958	966	949	938	(42)	
		Edh at Noon	65	76	87	104	107	106	112	107	90	72	63	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.18	3.09	4.41	5.64	6.58	7.64	7.21	6.34	5.29	3.71	2.45	1.79	(44)	
	RadStd		0.16	0.25	0.33	0.33	0.44	0.45	0.29	0.22	0.35	0.23	0.21	0.15	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page